DESIGNING WAVES FOR SURFING IN VIRTUAL REALITY

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1. Introduction

Chronic back and pelvic pain is a disability that can come from injuries or aging, resulting in a dull aching pain in the affected area. Studies have shown that chronic pain can alter the brain and neural system to send out pain signals despite the body not being actively harmed. However, surfing as an activity takes such mental focus that it has been observed to give some relief from lasting chronic pain. This chronic pain relief is the main priority of the virtual reality surfing simulator. Therefore, the goal of this project was to make the simulator as realistic feeling and looking as possible in order to fully immerse the user.

Purpose:
Improve the realism of the scene to deeper immerse the user into the surfing simulator

- **Goal 1:** Add interesting and realistic visuals with an accompanying audio
- **Goal 2:** Change the overall shape of the wave to make it look more realistic and to make it more fun for the user to surf
- **Goal 3:** Improve the quality of life of the UI in Unity to make it easier to change the settings of the scene

2. Methods

Adding assets to the scene:

1. Add assets of choice from Unity Asset Store to project library where they can be dragged onto the scene and modified
2. Write C# script to give the asset specified behavior such as movement or animation if needed
3. Find a mp3 file online for each of the assets that should play sound
   a. Attach this sound to the desired asset
   b. Set the sound settings, like if the audio loops or is played at a specific point in the asset's behavior

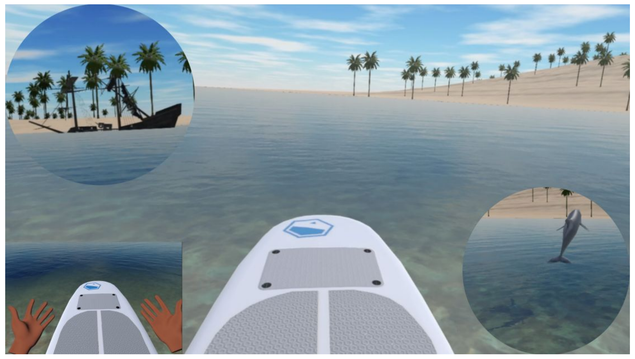


Figure 1: The virtual reality scene, some of the assets added (Eg. Dolphins, Trees, an Island, and a Shipwreck), and the motion tracking hands of the user.

Changing the shape and behavior of the wave:

1. Create a custom shader to create a bump on the ocean surface to increase the water height along the upper surface area of the shape it is attached to
2. Create a script to manage the dynamics as a result of adding the bump in water with tunable parameters like "Amplitude" which adjusts the height, "Time" in which the water is raised, "Period" or time between waves, "Delay" or the time after which the camera starts moving in order to consistently render the wave throughout the expected duration

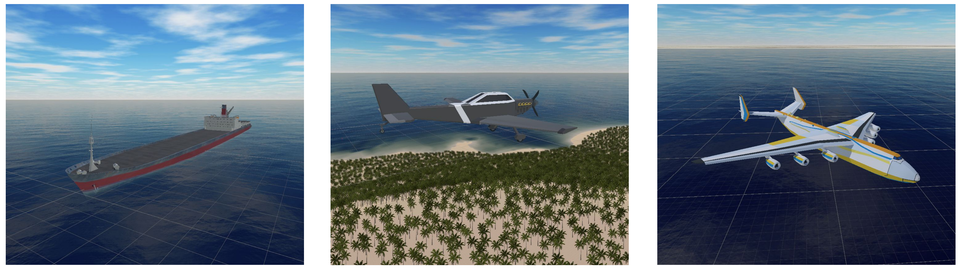
Improving the QoL of the Unity UI so settings can be easily adjustable:

1. Create a script that has buttons in the Unity UI (the Unity Inspector)
2. Set each button's function to move the observer (the individual inside the VR scene) to one of four places: where the waves start, near the shipwreck, on the beach, or on the dock

3. Contributions

Assets with an added behavior through the use of a script:

- The civilian transport airplane follows a path straight across the island, while the personal plane and the cargo ship both essentially circle the island. Spacial 3D audio was added for these three, as well as older assets like seagulls.



Assets that the user can interact with, but generally are not moving nor animated:

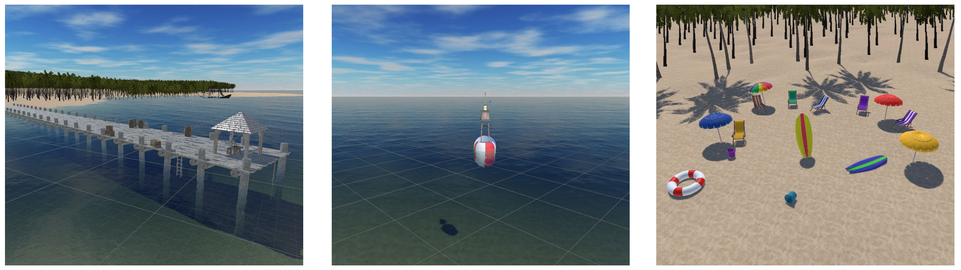


Figure 2: Screenshots of each of the added Assets.

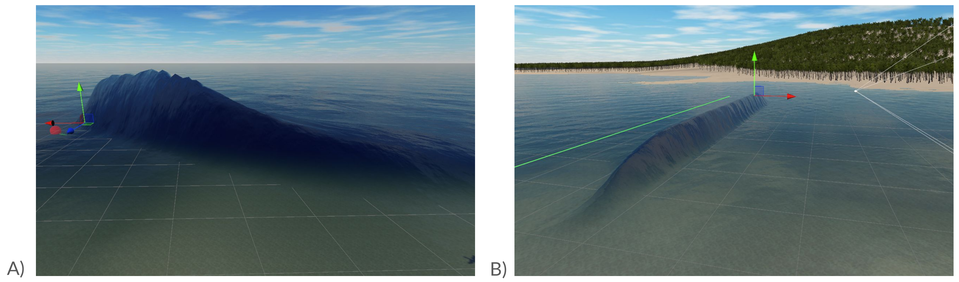


Figure 3: Screenshots of the old wave (A) next to the new wave (B). While the old wave raises the water at a point and then moves that point, the new wave raises a wall of water and then drops it immediately after to propagate a singular wave.

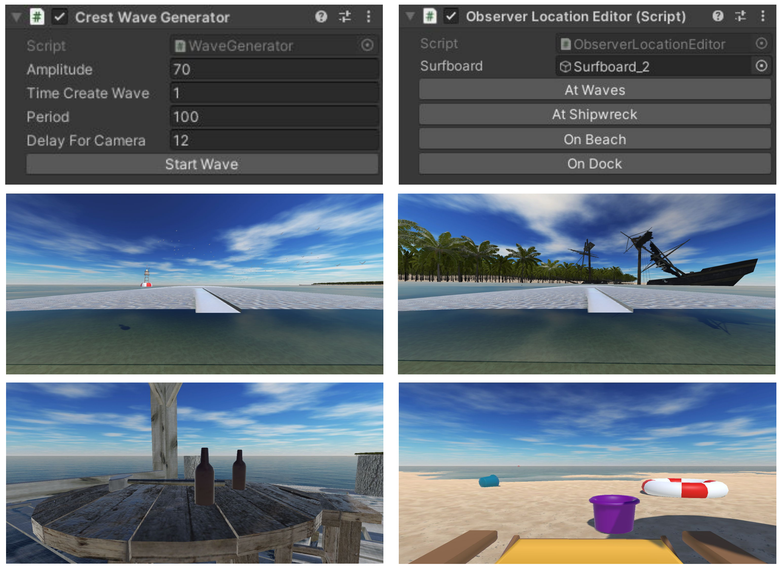


Figure 4: In the top left screenshot is of the "Wave Generator" script that allows for the customization of how the wave works and looks. The top right screenshot is a script that makes use of UI buttons in Unity to allow the operator to quickly and easily move the user in VR. Each of the four locations is also shown below the script from what would be the user's point of view.

4. Conclusion & Future Work

- **Goal 1:** Future work would see more randomized behavior of any moving assets. For example, the civilian transport plane or cargo ship starting at different points around the island or having a more randomized path.
- **Goal 2:** Since the desired shape is a smoother and more controlled wave, next steps will include systematic testing to see if the shape and behavior of the wave can be improved.
- **Goal 3:** This customization allows for new user studies to be done, such as what a patient feels like while on the water versus while sitting on a chair on the beach.
  - Adjustability would make setting up the scene more straightforward and user friendly.
    - Ex: including or excluding specific visuals/sounds/feedback, setting a specific time of day, or possibly even changing locations.

5. Acknowledgements

- Banerjee, P., Cherin, J., Upadhyay, J., Kutch, J., & Culbertson, H. (2024). Perception and Control of Surfing in Virtual Reality using a 6-DoF Motion Platform. arXiv preprint arXiv:2403.15924.

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